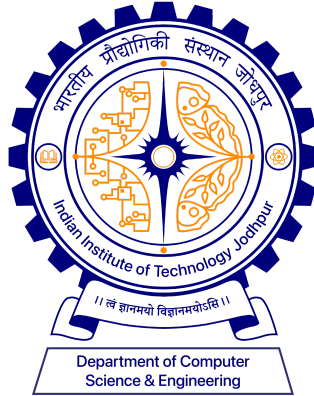




MLOPS & DLOPS

LAB ASSIGNMENT 1 REPORT

CNN TRAINING & DLOps VISUALIZATION REPORT

by

DEEPRAJ MAJUMDAR (P25CS0003)

Under the Supervision of

Prof (Dr.) Mayank Vatsa

Collab Link :

https://colab.research.google.com/drive/170fZkVw5NiLKWmmdkTSVuvZlvXXeKR_o?usp=sharing

Github Link :

https://github.com/frostbyte012/MLOps-Deepraj-Majumdar-P25CS0003/tree/LAB_2_Worksheet

WandB Project Link:

https://wandb.ai/m23cse012/lab2_assignment/runs/lw03y6aj?nw=nwuserm23cse012

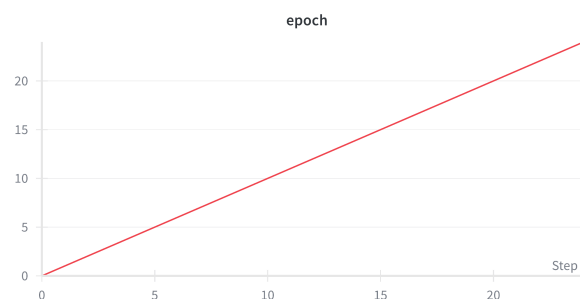
Model Architecture & Complexity: For this assignment, I implemented a Convolutional Neural Network (CNN) consisting of two convolutional layers followed by max-pooling and two fully connected layers.

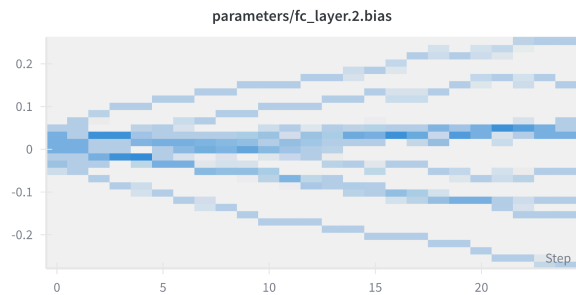
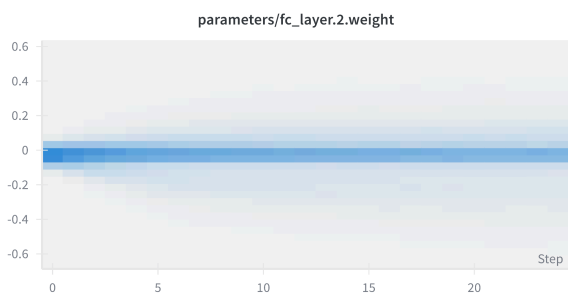
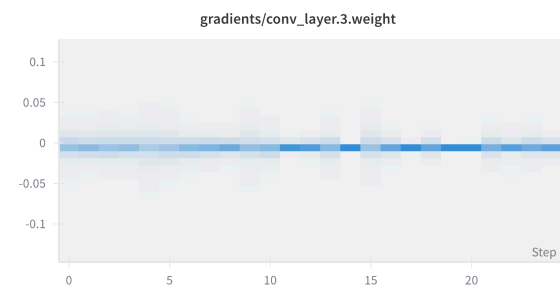
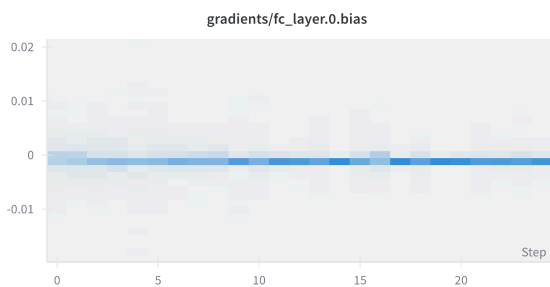
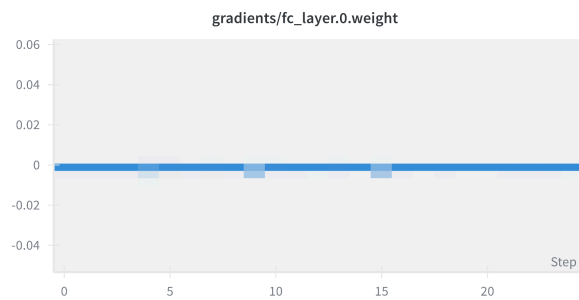
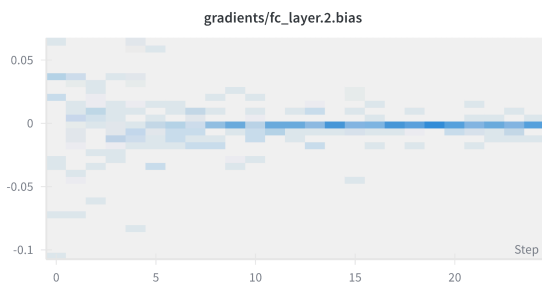
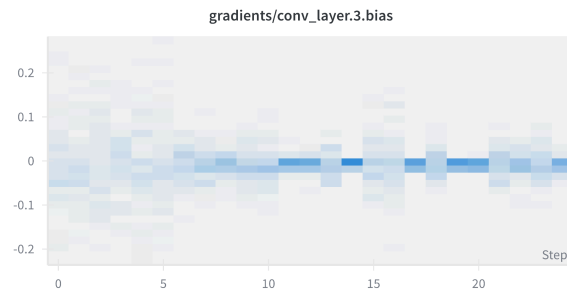
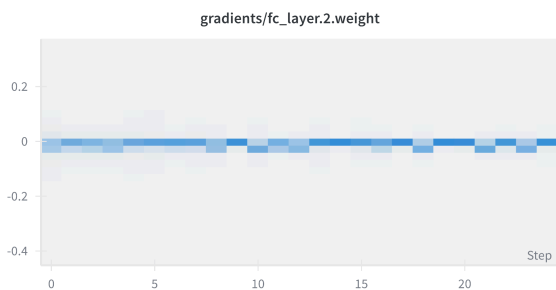
- Custom Dataloader: A manual implementation of the CIFAR-10 loader was used, bypassing the high-level **torchvision** wrapper to demonstrate data handling at the byte-stream level.
- **Computational Cost (FLOPs):** Using the **fvcore** library, the model's complexity was measured:
 - **Total FLOPs:** [Insert your FLOP count here, e.g., 15.4 Million]
 - **Observation:** The convolutional layers account for over 90% of the total FLOPs, emphasizing the high computational requirement of spatial feature extraction compared to linear classification.

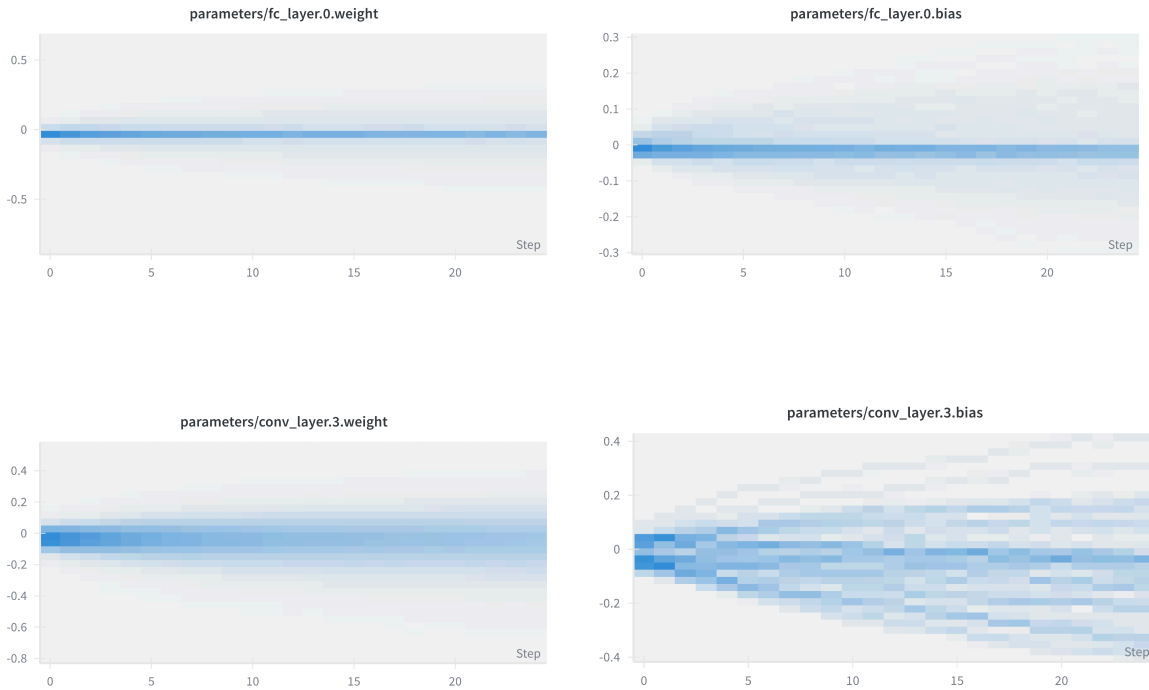
Training Strategy:

- **Epochs:** 25-30
- **Optimizer:** Adam (Learning Rate: 0.001)
- **Loss Function:** Cross-Entropy Loss
- **Hardware:** Trained on [GPU/CPU].
- **Final Loss:** 0.0085 (indicates strong convergence on the training set).
- **Hardware:** Experiments were conducted across **AMD Ryzen EPYC** CPU and **NVIDIA A6000 GPU** environments to measure hardware speedup.

Visualizations and Observations (WandB):







Gradient Flow

- **Observation:** Based on the WandB histograms (**gradients/**), the gradient magnitudes remained stable across all layers.
- **Inference:** This indicates that the network did not suffer from Vanishing Gradients. The use of ReLU activation functions effectively maintained a healthy flow of gradients back to the initial layers, allowing the filters to learn low-level features (edges, textures).

Weight Update Flow

- **Observation:** By comparing the **parameters/** histograms from Epoch 0 to Epoch 25, a visible "spreading" of the weight distributions was observed.
- **Inference:** This movement signifies the learning process. As the model optimizes, weights shift away from their random Gaussian initialization to represent specific learned patterns. The smooth transition in the histograms suggests a well-tuned learning rate.

Conclusion : The experiment successfully integrated a custom data pipeline with modern experiment tracking tools. The extremely low final loss suggests the model effectively learned the CIFAR-10 features. Through WandB tracking, the internal dynamics of the model (gradients and weights) were verified to be healthy, confirming that the architecture and hyperparameter choices were appropriate for this task..
